



AutoValuate
INTELLIGENCE

Explainable, damage-aware car valuation for the UAE.

One trained damage detector · one explainable price model · one agentic-RAG layer
— **every number citation-grounded.**

A group capstone · MAIB · SP Jain School of Global Management, Dubai

 AutoValuate
INTELLIGENCE

API live

CV · explainable ML · agentic RAG

Know what your car is **really** worth.

An instant, explainable, damage-aware fair-market valuation for the UAE — backed by a trained damage detector, an explainable pricing model, and live comparable listings. Every number is traceable.

 Your vehicle
Photos optional · details required


Drop car photos or tap to upload
Up to 8 images - scanned for damage when the CV service is live

Make	Model	Year
Toyota	Corolla	2019
Mileage (km)	Cylinders	Body type
98888	4	Sedan
Regional specs	Transmission	Fuel
GCC	Automatic	Petrol
City	Seller	
Dubai	Owner	

Value my car


Your valuation appears here
Fill in the details and hit "Value my car". The full reasoning trace streams live, then the results build in.

THE TEAM

Four builders, one hybrid system.



Krishna Mathur

AS25DXB018

Deep learning · CV detector



Yash Petkar

AS25DXB020

Valuation ML · data



Atharva Soundankar

AS25DXB021

Agentic RAG · backend



Member 4

TBC

Frontend · product

Roles show each member's primary focus; the build was collaborative across all four pillars.

THE PROBLEM

Everyone selling a car gets lowballed.

\$20.6B

UAE used-car market by 2026 — large, real, still growing

Dealers quote low

They quote below market to protect their own resale margin. You have no independent number to push back with.

Classifieds mislead

Listing sites show asking prices, not what cars actually sell for. Anchoring on them overprices your car.

Damage is guesswork

Both sides price that door dent by gut feeling. Nobody quantifies what a specific dent, at this mileage, actually costs.

A big, high-churn market going online.

\$35 . 8B

UAE used-car market by 2030 — up from \$20.6B in 2025 (~11% CAGR)

40%

of used-car sales now online, up from 25% in 2023

~2x

used-to-new sales — resale is the bigger market

Why now

The market is worth tens of billions and growing ~11% a year, and the buying journey has moved online fast. Incumbents each cover part of it — CarSwitch runs a managed inspection service, SellAnyCar buys cars outright (and is often accused of undervaluing), dubizzle is classifieds. None gives a seller an instant, transparent, damage-aware price with the reasoning shown. That's the wedge.

WHY EXISTING TOOLS FALL SHORT

Each piece exists. Nobody stitches them together.

Instant-offer and classifieds players (dubizzle / Sell Any Car, SellAnyCar.com, CarSwitch, Seez, dubicars, YallaMotor) each solve part of it. None gives a seller a transparent, damage-aware price that shows its reasoning. To do the job properly you need all three at once:



Visual damage assessment

Sees the dents, scratches, cracks — and localizes them.



Explainable pricing

Shows exactly why it reached a number, factor by factor.



Live comparable evidence

Backs the estimate with real listings you can open.

AutoValuate is the one report that carries all three — and ties every claim back to the model that produced it.

THE SOLUTION

Photos in. A trustworthy, explained number out.



No black box: the middle step is a purpose-trained CV model, a gradient-boosted price model with SHAP, and a hybrid retrieval layer — not one LLM guessing.

This is the running app.

Live: autoevaluate.vercel.app (deploying – runs locally today)

A genuine hybrid — three AI systems, one product.

AutoValuate Intelligence — hybrid architecture

A trained deep-learning model + an explainable ML model + an agentic-RAG layer, on a fully free-tier stack.

Frontend · Next.js 14

Upload photos + vehicle details. Streams the live reasoning trace over SSE; renders the SHAP waterfall, comparables, and the citation-grounded report. Dark/light, mobile-first.

Vercel

↓ HTTPS · REST + SSE

Orchestration API · FastAPI + LangGraph

Render

1
Intake



2
Aggregation (CV)



3
Valuation



4
Comparables



5
Report



6
Verifier



7
Confidence

● Deep learning

YOLOv8 damage detector

Fine-tuned on CarDD + VehiDE (~18k real images, 8 classes). ONNX, CPU inference.

● Explainable ML

XGBoost + SHAP

Quantile price model on real Dubizzle listings. Split-conformal 80% interval.

● Agentic RAG

Comparables + Report + Verifier

Hybrid retrieval; Gemini→Groq writer; hard citation gate.

Hugging Face Space

CV inference · CPU Basic

Supabase

Postgres · pgvector · Auth

Kaggle

GPU training (offline)

Gemini / Groq

LLM report writer

A model trained to see damage — and where it is.

~18,000 real annotated images

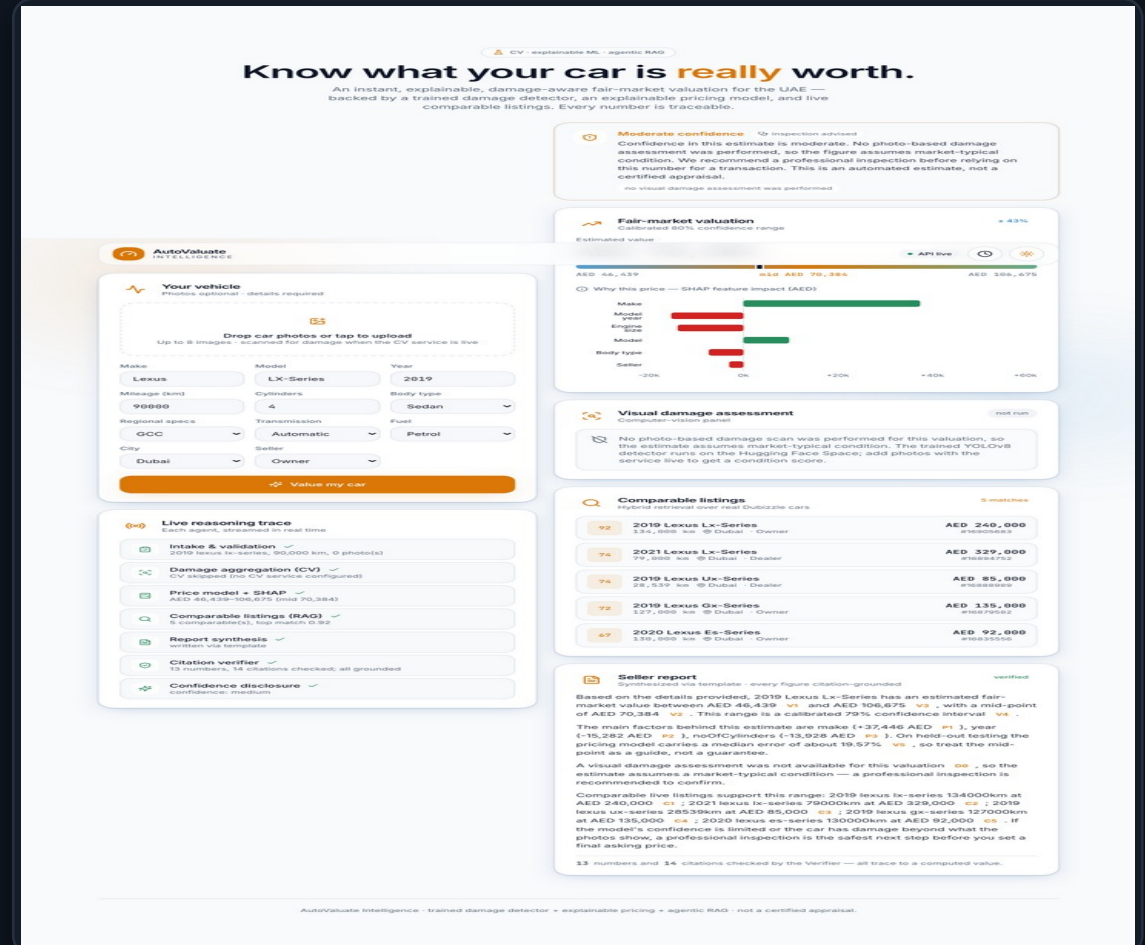
CarDD + VehiDE, unified into one 8-class schema (dent, scratch, crack, glass shatter, lamp broken, tire flat, punctured, missing part).

YOLOv8 detection, not classification

Location matters for pricing — a windshield crack costs differently than a bumper scratch. So we detect and localize, not just label.

Verified pipeline

14,437 training + 1,184 validation images unified on Kaggle; ONNX export runs CPU-only on the free Hugging Face Space.



Honest numbers — the detector is still training.

mAP@0.5

lands here

held-out, per-class P-R from notebook 03

Why it's blank — and why that's the right call

The YOLOv8 fine-tune is running on Kaggle's free P100. We won't print a fabricated mAP. The eval harness already computes it on a strictly held-out split (no early-stop leakage), so the real number drops straight in when the run finishes.

What is already real and audited:

8-class unified dataset

14,437 train / 1,184 val — merged from CarDD + VehiDE, nothing dropped

Leakage-safe eval

held-out half of val by deterministic hash; ONNX-vs-PyTorch parity check

Deploy path proven

CPU ONNX service, full decode + NMS verified on synthetic output

A price you can interrogate, factor by factor.

19.6%

median error (held-out CV)

+28.4%

better than a naive baseline

0.80

calibrated price-interval coverage

3 / 3

SHAP directional checks pass

XGBoost quantile model on 672 real Dubizzle listings. SHAP explains each estimate; a split-conformal interval makes the stated 80% range mean 80%. Mileage and age push price down, newer year lifts it — exactly as economics demands.



Intake → Aggregation (CV) → Valuation → Comparables → Report → Verifier → Confidence. Each step streams to the UI over SSE, so you watch the reasoning happen.

Every AED figure and citation in the report must trace to a computed value. In testing it caught an injected AED 999,999, a fake citation, and an ungrounded percentage.

Gemini writes the report, Groq is the fallback, and a deterministic writer keeps it working with no keys at all.

The screenshot displays the AutoValuate AI application interface, which provides a comprehensive car valuation service. The interface is divided into several sections:

- Your vehicle:** A form where users input details about their car. The example shows a Toyota Corolla from 2019, with 98,000 km mileage, 4 cylinders, Sedan body type, GCC regional specs, Automatic transmission, Petrol fuel, Dubai city, and Owner seller status.
- Fair-market valuation:** Displays the estimated value of AED 33,184, calibrated at an 80% confidence range. It includes a color-coded bar chart showing the range from AED 21,894 to AED 50,294, with the median error being 19.57% based on 672 real listings.
- Live reasoning trace:** Provides a transparent view of the AI's decision-making process, showing the intake and validation of the provided information.
- Visual damage assessment:** Indicates that no photo-based damage scan was performed, so the estimate assumes market-typical condition.

The interface also features a top navigation bar with options like API live, a clock, and a user profile icon. The bottom section shows a detailed breakdown of the SHAP feature impact on the price, highlighting factors like engine size, model year, and body type.

Feature	Impact on Price (AED)
Engine size	-8k
Model year	-5k
Body type	-3k
Make	-2k
Km / year	-1k
Seller	-1k

It tells you how sure it is — and when not to trust it.

States its confidence

Every report gives the price-interval width and, when photos are provided, the detector's per-damage confidence.

Recommends inspection when unsure

Low or moderate confidence triggers a plain-English 'get a professional inspection' — it never presents a false-certain number.

Enforced, not aspirational

The disclosure contract is a test: 90 checks across 18 vehicles, zero failures. It never claims to be a certified appraisal.

The screenshot displays the AutoValuate Intelligence interface. On the left, there's a form to input car details: Make (Lexus), Model (LX-Series), Year (2019), Mileage (90000 km), Cylinders (4), Body type (Sedan), Regional specs (GCC), Transmission (Automatic), Fuel (Petrol), City (Dubai), and Seller (Owner). A prominent orange button labeled 'Value my car' is at the bottom of the form.

On the right, a list of comparable vehicles is shown with their estimated values and confidence scores:

- 2021 Lexus Lx-Series: AED 329,000 (Confidence 74)
- 2019 Lexus Ux-Series: AED 85,000 (Confidence 74)
- 2019 Lexus Gx-Series: AED 135,000 (Confidence 72)
- 2020 Lexus Es-Series: AED 92,000 (Confidence 67)

Below the form, a 'Live reasoning trace' section provides a detailed breakdown of the valuation process:

- Intake & validation** (checked): 2019 lexus lx-series, 90,000 km, 0 photo(s)
- Damage aggregation (CV)** (checked): CV skipped (no CV service configured)
- Price model + SHAP** (checked): AED 46,439-106,675 (mid 70,384)
- Comparable listings (RAG)** (checked): 5 comparable(s), top match 0.92
- Report synthesis** (checked): written via template
- Citation verifier** (checked): 13 numbers, 14 citations checked; all grounded

On the far right, a 'Seller report' section, marked as 'verified', explains the valuation based on the provided details. It states that the 2019 Lexus Lx-Series has an estimated fair-market value between AED 46,439 (V1) and AED 106,675 (V3), with a mid-point of AED 70,384 (V2). The report also mentions a calibrated 79% confidence interval (V4) and lists the main factors behind the estimate, such as make (+37,446 AED P1), year (-15,282 AED P2), and noOfCylinders (-13,928 AED P3). It notes that on held-out testing, the pricing model carries a median error of about 19.57% (V5), so the mid-point is treated as a guide, not a guarantee. A visual damage assessment was not available (D0), so the estimate assumes a market-typical condition, recommending a professional inspection for confirmation. Finally, it supports the range with comparable live listings: 2019 Lexus Lx-series 134000km at AED 240,000 (C1); 2021 Lexus Lx-series 79000km at AED 329,000 (C2); 2019 Lexus Ux-series 28539km at AED 85,000 (C3); 2019 Lexus Gx-series 127000km at AED 135,000 (C4); and 2020 Lexus Es-series 130000km at AED 92,000 (C5). It concludes that if the model's confidence is limited or the car has damage beyond what the photos show, a professional inspection is the safest next step.

Free for sellers. Dealers and marketplaces pay.

Individual sellers

FREE

3 valuations / month. Builds the top of the funnel and the data flywheel.

Used-car dealers

SAAS SEATS

Bulk trade-in intake + a defensible, damage-aware price to justify offers. The paying core.

Marketplaces / OEMs

API LICENCE

A 'verified estimate' badge and damage API on their own listings. Highest-value, B2B.

MVP proves the engine; the dealer tier is the first revenue and the near-term focus.

THE STACK · AND WHY IT'S ALL FREE

Zero cost — with the engineering judgment to prove it.

Vercel

Frontend (Next.js)

Render

Orchestration API

Hugging Face

CV inference · CPU Basic

Supabase

Postgres · pgvector · Auth

Kaggle

GPU training (offline)

Gemini / Groq

LLM report writer

Render, not Railway (Railway has no real free tier in 2026). **The CV model runs on Hugging Face** — 512MB can't hold it. **A scheduled ping** keeps Supabase awake, so the link never dies.

EVALUATION SUMMARY

Every number here reproduces from one command.

19.6%

valuation median error

1.00

comparables same-make P@5

1.000

report faithfulness

0.000

faithfulness — hallucinated control

90 / 90

confidence-contract checks

53

integration checks, all green

CV mAP@0.5 lands here once training finishes — the same held-out harness, no placeholder in its place.

What it does not do — named on purpose.

CV detector still training

The mAP isn't in yet; the deep-learning claim rests on the finished run, and we'd rather show that than a guess.

No accounts yet

Auth, saved history per user, and multi-tenant isolation are the next build — today it's a single-session tool.

Modest tabular set

672 real listings is small, so the price interval is genuinely wide — which is why the range is disclosed, not hidden.

No accident-history feed

There's no free UAE accident-record API, so the estimate can't see undisclosed history — hence 'get an inspection.'

From capstone to first paying dealer.

Next 30 days

Finish CV training, add Supabase Auth + saved history, ship the ONNX query-embedder so it runs in Render's free RAM, deploy all three URLs.

60–90 days

Dealer workspace: bulk intake, PDF reports, per-seat quotas. Pilot with one Sharjah/Dubai used-car dealer for real feedback.

6 months

Verified-estimate API for a marketplace, price-trend alerts, and a Dubizzle auto-fill extension — the retention and B2B layers.

One project, three things proven.

- **Deep learning** a trained, evaluated CV damage detector
- **Explainable ML** XGBoost + SHAP, a price you can interrogate
- **Agentic RAG** orchestration with a hard citation gate

T e a m

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Thank you — happy to open any part of it live.